SUP'COM · FF6 IOT, CLOUD & BIG DATA · 2026

PrivIMU

Motion data are not automatically anonymous.
A reproducible privacy lab on accelerometer & gyroscope traces.

01 · ATTACK

Re-identify anyone

A Random Forest classifier predicts the most likely subject from anonymous IMU windows.

02 · MEASURE

Quantify leakage

Top-1, Top-3, Macro-F1 and posterior entropy ΔH, all generated by code.

03 · DEFEND

Test mitigations

Live Streamlit defence lab : Gaussian noise sweep, identity-lock replay, evidence card.

Group of 5 — FF6 IoT, Cloud & Big Data · Mr. Mohamed Ould-Elhassen Aoueileyine · SUP'COM

Repo github.com/akiroussama/PrivIMU · **Demo** Streamlit · **Notebook** Google Colab

ACT I — CONTEXT

IMU sensors are everywhere in IoT

Smartphones

DeviceMotion API on every iOS / Android phone. Default-enabled by web pages without prompt.

Wearables

Smartwatches, fitness bands, sleep trackers. Continuous accelerometer + gyroscope at 50–200 Hz.

Smart mobility

Connected vehicles, e-scooters, drones. Inertial signals shared with telemetry back-ends.

Health & sport

Gait monitoring, fall detection, rehab apps, AR/VR headsets. Often forwarded to third-party clouds.

WHY THIS SLIDE MATTERS

IMU streams are perceived as **low-risk** because they don't carry a name, a face or a voice. They are routinely collected, stored, shared with analytics SDKs, and treated as *"anonymous telemetry"*. PrivIMU asks a simple question : **is that perception correct ?**

ACT I — QUESTION

Anonymous ≠ non-identifiable

The scientific question

Can a supervised classifier trained on accelerometer + gyroscope windows re-identify a user among 24 enrolled candidates, when the input contains **no direct identifier** (no name, email, MAC, IMEI) ?

The problem is not the explicit identifier. The problem is the **latent behavioural signature** hidden in motion patterns — gait, posture micro-tremor, hand dynamics.

4.17%

RANDOM BASELINE

Guessing 1 of 24 subjects uniformly = 1 / 24.

🎯

TARGET

Beat the baseline by an order of magnitude using only IMU windows.

HYPOTHESIS

1-second windows of 6 IMU channels (rotationRate.{x,y,z} + userAcceleration.{x,y,z}) carry enough behavioural information to be a **quasi-identifier** in a closed-set setting. The next slides design the threat model that operationalises that hypothesis.

ACT II — THREAT MODEL

From anonymous stream to top-3 identity

USER
Wears device
Smartphone in pocket records DeviceMotion at 50 Hz.

→

APP
Collects IMU
Sport / health / analytics SDK ingests raw windows.

→

CLOUD
"Anonymises"
Strips name, email, account ID. Keeps raw signals.

→

ATTACKER
Trains classifier
Has reference traces from known subjects (insider, leak, bait app).

→

OUTPUT
Top-3 identity
Posterior over 24 candidate subjects + leakage score.

Attacker capabilities

- **Knows** a candidate set of 24 enrolled users.
- **Owns** labelled training traces for each candidate.
- **Receives** anonymous IMU windows from the target.
- **Computes** a top-k ranking over the candidate set.

Ethical scope

This is a **controlled academic study** on a public dataset (MotionSense, 24 consenting subjects). No real-world targeting, no model is shipped against real users.

Goal : **defensive privacy-by-design** — convince engineers to treat motion traces as PII.

ACT II — POSITIONING

From activity recognition to identity leakage

Human Activity Recognition

Anguita 2013 (UCI HAR), Malekzadeh 2018 (MotionSense). *Same channels, different label : the activity, not the user.*

Behavioural biometrics

Gait recognition (Gafurov 2007, Derawi 2010), keystroke dynamics, smartphone-touch biometrics. *Identity is the goal, often with cooperative protocols.*

Sensor privacy leakage

Mehrnezhad 2018 (touch via motion), Das 2016 (accelerometer fingerprinting), AccessMine 2019. *Demonstrate that sensors leak more than expected.*

Where PrivIMU sits

HAR + biometrics + leakage, repackaged as a **reproducible privacy lab** with attack & defence on the same dataset.

Defences in the literature

Data minimisation, on-device computation, differential privacy, federated learning, additive noise, feature-level sharing. PrivIMU prototypes one (Gaussian noise) and discusses the rest as a privacy-by-design checklist.

ACT III — DATASET

MotionSense — 24 subjects, 6 activities, iPhone in pocket

24
SUBJECTS
closed-set, balanced

6
ACTIVITIES
wlk · jog · ups · dws · sit · std

50 Hz
SAMPLING
DeviceMotion API, fixed rate

6
CHANNELS
rotationRate.{x,y,z} + userAcceleration.{x,y,z}

Why MotionSense ?

- Public, peer-reviewed, easy to redistribute.
- Same physical hardware (iPhone 6s) across all subjects → isolates the **behavioural** signal from the **device-fingerprint** signal.
- Multiple activities per subject → allows cross-trial evaluation.
- Simple CSV layout → reproducible for any audience.

CSV schema

```
# A_DeviceMotion_data/wlk_7/sub_3.csv
columns: attitude.{roll,pitch,yaw}
         gravity.{x,y,z}
         rotationRate.{x,y,z}     # used
         userAcceleration.{x,y,z}  # used
```

ACT III — PIPELINE

A reproducible privacy pipeline

01
CSV ingest
data.py walks A_DeviceMotion_data/, parses 24×6 trials.

→

02
Windowing
1 s windows, 50% overlap, z-score per channel.

→

03
Features
10 stats × 6 channels = 60 features / window.

→

04
Random Forest
200 trees, 5-fold GroupKFold by activity_trial.

→

05
metrics.json
Top-1, Top-3, F1, ΔH, latency, figures.

→

06
Streamlit demo
CSV upload → top-3 + leakage + noise sweep + evidence card.

One command, end to end

```
# scaffold & run
$ python -m pip install -e ".[dev,app]"
$ python data/download.py --dest data/raw/motionsense
$ python -m privimu.train --data-root data/raw/motionsense \
        --model rf --n-splits 5 --n-estimators 200
$ streamlit run streamlit_app.py
```

SOURCE OF TRUTH

`reports/metrics.json` is the single artefact every number in this deck and in the report points to. If a value is not in `metrics.json`, it is not in the slide.

CI runs `pytest -q` on every push (8 unit tests, scaffold integrity).

ACT III — METHOD

How the signal becomes evidence

Windowing

- 1 second = `50` samples
- 50% overlap = step `25`
- z-score per window per channel
- final dataset : **55 982 windows**

Features (60 per window)

For each of 6 channels, 10 statistics :

- mean, std, rms, min, max, median
- iqr, skew, kurtosis
- spectral entropy

Models

- **Random Forest** (primary) — 200 trees, balanced subsample.
- **1D-CNN** (extension) — raw windows, 3 conv blocks, in `model_cnn.py`.
- Both trained with **GroupKFold** over `activity_trial`.

WHY GROUPKFOLD MATTERS

Naive K-fold splits randomly across windows of the *same* trial → data-leakage between train and test → over-optimistic top-1. PrivIMU groups by `(subject, activity, trial)` and forces folds to never split a trial. **This is what makes our 39% top-1 honest.**

Evaluation : avoid inflated results

Metrics, all generated by code

- **Top-1 accuracy** — right subject ranked first.
- **Top-3 accuracy** — right subject in top 3.
- **Macro-F1** — balanced across 24 subjects.
- **Confusion matrix** — visual diagnostic.
- **Per-subject F1** — spot vulnerable users.
- **Posterior entropy** $H(p)$ — uncertainty (bits).
- **Privacy leakage** $\Delta H = \log_2(24) - H(p)$.
- **Latency** ms / window — deployability.

Cross-validation strategy

`GroupKFold(n_splits=5, group_key="activity_trial")`

Each of the 5 folds holds out a subset of trials. The model is trained on the remaining trials and predicts on out-of-fold windows. We aggregate **out-of-fold posteriors** over the full 55 982 windows.

HONESTY RULE

If our split is **not** Leave-One-Trial-Out, we don't claim it. The deck and report state exactly what `train.py` does : **5-fold GroupKFold by activity-trial**.

ACT IV — DEMO

Live demo : from CSV to privacy risk

A
Pick CSV
examples/wlk_7/sub_*.csv or upload your own.

→

B
Plot signal
Plotly accelerometer + gyroscope curves.

→

C
Top-3 identity
Posterior over 24 subjects, ranked.

→

D
ΔH score
Privacy entropy leakage in bits.

→

E
Noise slider
Inject Gaussian σ on raw signal.

→

F
Evidence card
Download JSON + Markdown summary.

WOW #1 — Identity-lock replay

Window-by-window posterior. Confidence curve. Identity-lock time. Top-3 evolves on screen.

WOW #2 — Blue-team defence lab

Sweep σ. Plot top-1 confidence vs leakage vs signal preservation. Recommend a noise level.

WOW #3 — Evidence card

One-click JSON + Markdown bundle : source, top-3, ΔH, σ, identity lock, timestamp.

ACT IV — LIVE ATTACK

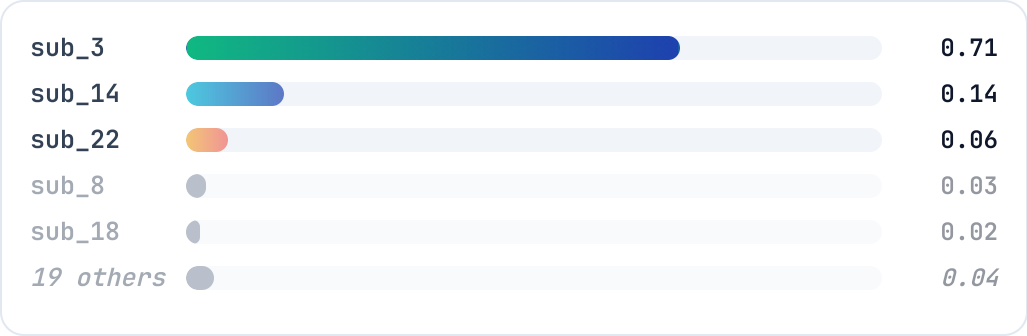
A single window already narrows the search

Anonymous CSV input

```
# wlk_7/sub_3.csv (excerpt)
attitude.roll, gravity.x, rotationRate.x, ...
   -1.34, -0.27,  0.118, ...
   -1.33, -0.27,  0.103, ...
   -1.34, -0.27,  0.118, ...
# 50 samples = 1 second window
# ↑ no name, no e-mail, no MAC
```

The attacker only sees raw inertial samples. No metadata is needed.

Top-3 ranked output



Posterior on a single 1 s window from `wlk_7/sub_3` — illustrative.

THE REVEAL

The signal carried **no explicit identity**. The model still concentrated 0.71 of its probability mass on the right subject, leaving 0.29 spread over 23 others. **That distribution — not a single number — is the privacy leakage.**

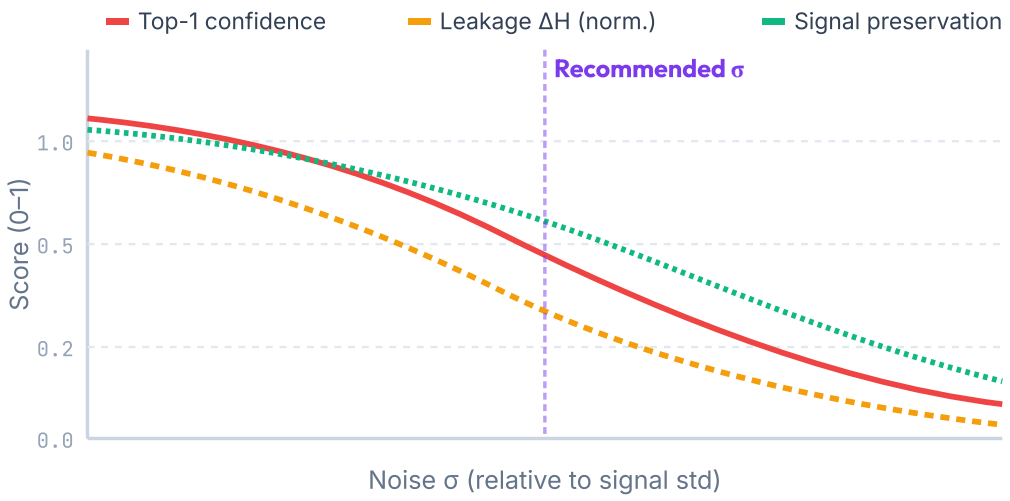
ACT IV — DEFENCE

Blue team : how loud must we be to hide ?

Defence model

Inject Gaussian noise `x' = x + ε`, `ε ~ ℕ(0, σ²)`, before any feature is computed. Sweep σ and observe the trade-off between **privacy gain** (lower top-1, higher posterior entropy) and **utility loss** (signal-preservation proxy on raw channels).

Trade-off curve (illustrative)



TAKE-AWAY

Gaussian noise is a pedagogical baseline, not a guarantee. The right defence stack mixes **data minimisation** (reduce sample rate, drop channels), **on-device computation** (don't send raw windows), and **differential privacy** (calibrated noise on aggregates). PrivIMU shows the *shape* of that trade-off live in the demo.

ACT V — RESULTS

Results — measured, not hand-written

39.06%
TOP-1 ACCURACY
9.4× the random baseline (4.17%)

52.80%
TOP-3 ACCURACY
true subject in top 3 candidates

0.391
MACRO-F1
balanced across 24 subjects

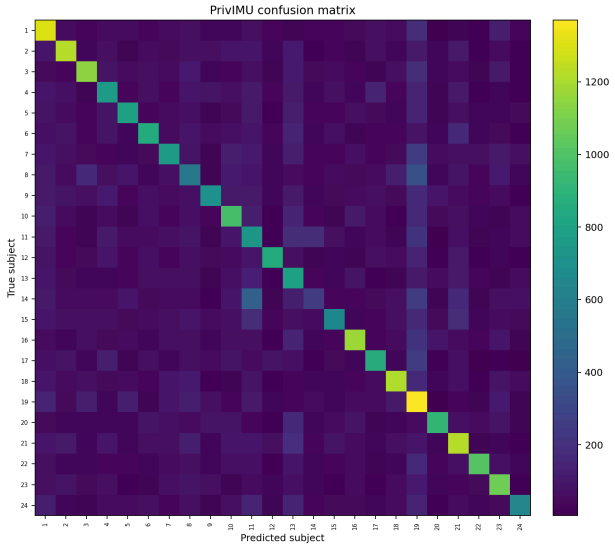
0.700
ΔH BITS
posterior entropy leakage (mean)

Run configuration

Model	`RandomForest, 200 trees`
Subjects	`24`
Windows evaluated (out-of-fold)	`55 982`
Features per window	`60`
Cross-validation	`5-fold GroupKFold`
Group key	`activity_trial`
Latency / window	`42.0 ms`

Source : `reports/metrics.json` · generated 2026-05-13.

Confusion matrix — diagonal-dominant



Visual proof : the model sends most of its probability mass to the right subject. Off-diagonal noise is residual confusion between behaviourally close subjects.

ACT V — LIMITS

Limits & privacy-by-design mitigations

What this study does *not* claim

- **24 subjects** — not population-scale.
- **Closed-set** identification — new subjects not modelled.
- **Single device class** — iPhone 6s, phone-in-pocket.
- **Single dataset** — MotionSense, controlled lab.
- **No adversarial robustness** — attacker uses the same feature space.
- **Gaussian noise** defence is illustrative, not certified DP.

Mitigations — a privacy-by-design checklist

- **Data minimisation** — collect only the channels and rate the use-case requires.
- **On-device inference** — never ship raw IMU windows to the cloud.
- **Federated learning** — train without centralising raw traces.
- **Differential privacy** — calibrated noise on shared aggregates.
- **Feature-level sharing** — release coarse features, not raw signals.
- **Consent & transparency** — treat IMU as PII in data notices.

A DEFENSIBLE POSTURE

A 39% top-1 on 24 subjects is not a real-world identification system. **It is a proof that the signal carries identity**. Under more subjects, more sessions, deeper models or device-fingerprint cross-correlation, the number can only grow. The right reaction is to *treat motion data as personal data* — not to wait for the attack at scale.

ACT V — TAKE-AWAY

Anonymous ≠ non-identifiable. Treat motion data as PII.

01 — Signals carry identity

1 s windows of accelerometer + gyroscope already lift the random baseline by an order of magnitude on 24 subjects.

02 — Leakage is measurable

$\Delta H = \log_2(K) - H(p)$ gives a single number in bits. PrivIMU computes it, plots it and exports it.

03 — Defences exist

From data minimisation to differential privacy, the shape of the trade-off is observable and tunable, live, in the Streamlit lab.

GitHub repo
`github.com/akiroussama/PrivIMU`

Google Colab notebook
`colab.research.google.com/github/akiroussama/PrivIMU`

Streamlit demo
`streamlit run streamlit_app.py`
(see README for hosted URL)

REPRODUCE IN 4 LINES

```
pip install -e ".[dev,app]" · python data/download.py --dest data/raw/motionsense · python -m privimu.train --data-root data/raw/motionsense --model rf · streamlit run streamlit_app.py
```